

Types of Linked Lists

Q. Elaborate on the different types of Linked Lists. Explain Singly, Circular, and Doubly Linked Lists with their structures and diagrams.

[DEF] Definition to Remember

Linked Lists are classified based on how nodes are connected: Singly Linked List (one-way traversal), Circular Linked List (last node points to first, continuous circle), and Doubly Linked List (two-way traversal using previous and next pointers).

1. Singly Linked List

The simplest type. Each node contains data and a single pointer (next) pointing to the next node.

Diagram:

2. Circular Linked List

The next pointer of the last node does NOT point to NULL. Instead, it points back to the first node (head), forming a continuous circle.

Diagram:

3. Doubly Linked List

Each node contains three fields: data, a pointer to the next node, and a pointer to the previous node.

Structure in C:

Diagram:

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Three primary types: Singly, Circular, and Doubly Linked Lists.
2. Singly Linked List: contains data and next pointer; unidirectional traversal; ends with NULL.
3. Circular Linked List: last node points back to the head; continuous traversal; no NULL at the end.
4. Circular lists are used in Round-Robin CPU scheduling.
5. Doubly Linked List: contains prev, data, and next pointers; bidirectional traversal.
6. Doubly lists use more memory per node but allow moving backwards easily.
7. Doubly lists are used in browser history (back/forward functionality).

[RECALL] Quick Recall

Singly (Forward only, ends in NULL) -> Circular (Last points to Head, continuous loop) -> Doubly (prev + next pointers, bidirectional, uses more memory)